

TIC 403080487

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 2.290237 d
Epoch (t0) 1625.7736 BTJD
Duration 0.1700 d
Depth 0.808 % (8,076 ppm)
SNR 110.4

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

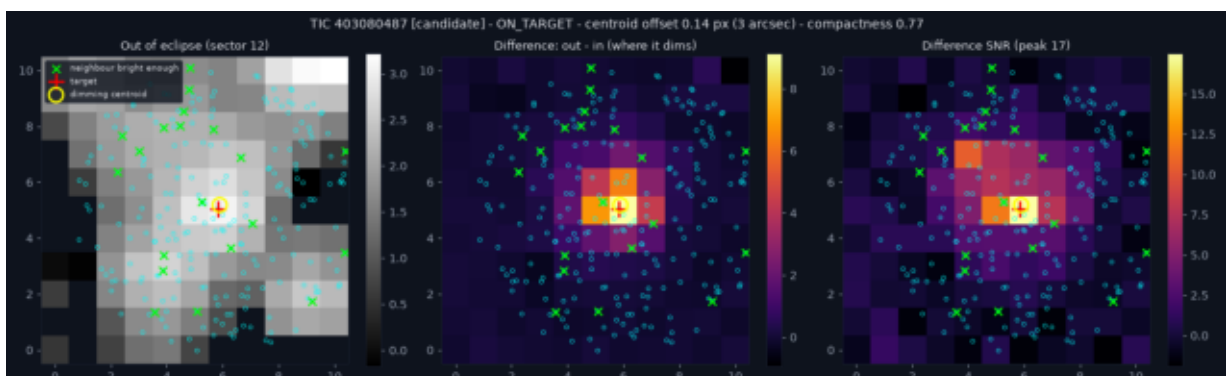
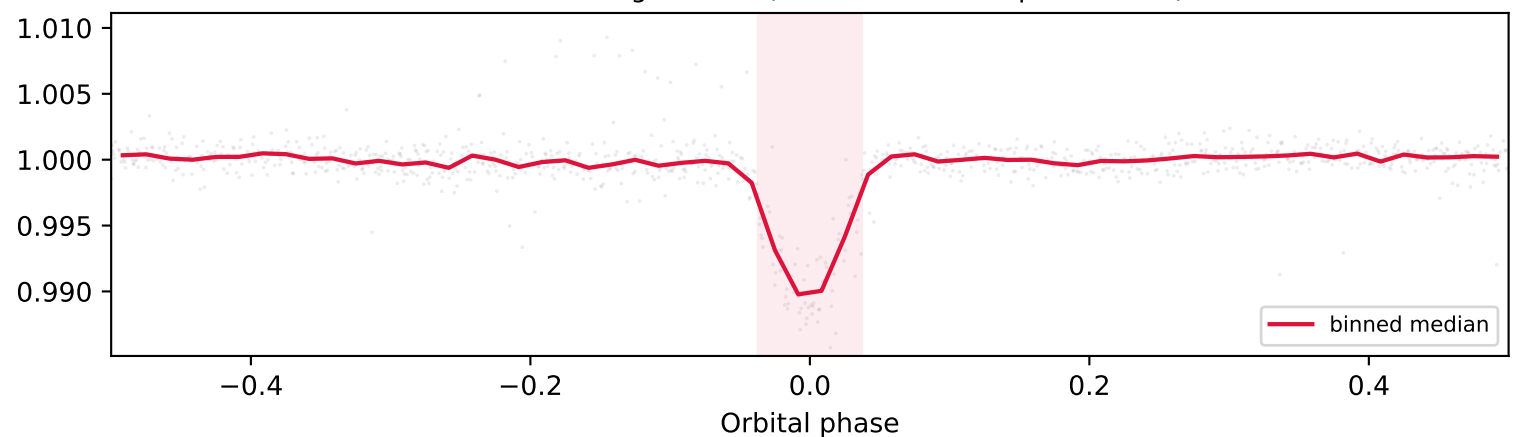
Eclipses observed 11 (gate ≥ 3)
Robust depth ratio 1.00 (gate ≥ 0.50)
Robust depth sigma 66.2 (gate ≥ 7.0)
Odd/even z-score 1.23 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 12 light-curve dip in this sector: 66.5 sigma t0 refined by 0.0075 phase
Difference-image peak SNR 17.49 (gate ≥ 5.0) Compactness 0.77 (gate ≥ 0.50)
Dimming centroid offset 0.14 px (3 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 10001763019
0.65 px away, dTmag +5.13 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.